



For an Open Mind

UGANDA TECHNOLOGY AND MANAGEMENT UNIVERSITY

UTAMU

**IS 200 - SYSTEMS ANALYSIS AND DESIGN
GROUP ASSIGNMENT COURSEWORK 2
YEAR 2 SEMESTER 1**

**A FEASIBILITY STUDY REPORT ON A
PROPOSED STUDENT MANAGEMENT SYSTEM (SMS)**

**FOR
WOMEN'S INSTITUTE OF TECHNOLOGY & INNOVATION
(WITI)**

PREPARED ON 14th/10/2024

BY

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1. EXECUTIVE SUMMARY

This feasibility study explores the practicability of developing and implementing a Student Management System (SMS) for Women's Institute of Technology and Innovation (WITI). The primary purpose of the SMS is to streamline administrative processes, enhance data accuracy and security, improve reporting capabilities, and support institutional growth. By addressing the current challenges of a manual system, the SMS aims to provide WITI with efficient management of student data and automated attendance tracking hence facilitating a more effective educational environment.

2. BACKGROUND

2.1 Organization Background

Women's Institute of Technology and Innovation (WITI) is a Code Academy founded by Women In Technology Uganda (WITU) with a strong vision to close the technological gender gap in Uganda. Located in Kamukya, Kanjokya Street, opposite Cavendish University, WITI focuses on empowering young women with the necessary skills to excel in the technology sector. The institute offers a Diploma in Computer Science and Entrepreneurship, catering to female students who have completed the Ugandan Advanced Certificate of Education (UACE) with a science-focused curriculum at their A-level.

2.2 Problem Statement

Women's Institute of Technology and Innovation (WITI) currently relies on file-based approaches for managing student records and paper-based methods for attendance tracking. This leads to duplication of records, frequent errors in record-keeping and attendance tracking and delays in accessing vital information. The absence of a centralized system hinders efficient data management and timely decision-making, affecting both students and staff.

2.3 Purpose for the Study

The purpose of this feasibility study is to evaluate the potential of implementing a Student Management System (SMS) at Women's Institute of Technology and Innovation (WITI).

Main Objective of the Study

- To design and implement a comprehensive Student Management System (SMS) WITI that digitizes students records management. which will replace the file-based approaches and paper-based methods with a centralized digital solution improving accuracy, eliminating data redundancy, ease accessibility and operational efficiency.

Specific Objectives

- To gather requirements for the proposed digital system.
- Design the process flows for the proposed system.
- Develop a digitized Student Management System.
- Test, Deploy and Implement the proposed system.

2.4 Scope of the Study

This feasibility study encompasses the following areas:

Technical Aspects: Evaluation of the technical requirements for implementing a Student Management System (SMS), including infrastructure, software, and hardware capabilities necessary for efficient operation.

Economic Aspects: Analysis of the financial implications of transitioning to a SMS, including initial costs, potential savings, and return on investment.

Operational Aspects: Examination of how the SMS will integrate into existing workflows, improve administrative processes, and enhance overall institutional efficiency.

3. DESCRIPTION OF THE PROPOSED SYSTEM

3.1 System Overview

The proposed Student Management System (SMS) will automate administrative processes at WITI, including:

- **Student Registration:** Streamlining the enrollment process for new students.
- **Report Generation:** Generating detailed performance and attendance reports.
- **Grading Management:** Automating grade entry and access for students and faculty.
- **Attendance Tracking:** Real-time recording and monitoring of student attendance.

3.2 System Features

Key features will include:

- **User-Friendly Interface:** Intuitive navigation for students and administrators.
- **Secure Data Management:** Protecting sensitive student information.
- **Real-Time Analytics:** Dashboards for tracking progress efficiently.
- **Integration Capabilities:** Smooth transition and data consistency with existing systems.
- **Mobile Accessibility:** Smooth access via mobile devices for convenience.

3.3 System Benefits

The anticipated system benefits for WITI include:

- **Increased Efficiency:** Reducing administrative workload and processing time.
- **Improved Data Accuracy:** Minimizing errors through automation.
- **Enhanced Student Monitoring:** Real-time tracking of attendance and performance.
- **Better Resource Management:** Efficient allocation of resources.
- **Support for Decision-Making:** Data-driven insights for informed institutional decisions.

4. FEASIBILITY ANALYSIS

4.1 Technical Feasibility

4.1.0 Current Infrastructure Assessment: WITI possesses a robust technological infrastructure, including:

- **Hardware:** Sufficient computers in student labs and laptops provided to staff, all equipped with necessary software (like, web browsing).
- **Network:** Reliable Wi-Fi access supported by a network server.
- **Cloud Hosting:** A cloud shared hosting server already in use for the institution's website, with good security protocols, including secure protocols and SSL certificates.
- **Operating Systems:** Both the computers in the computer and the laptops provided to the staff member have upto date Operating Systems (Windows, Ubuntu, MacOS)

Support for the Proposed SMS: The existing infrastructure is capable of supporting the proposed SMS, as all necessary components are in place. The cloud server can accommodate the SMS, ensuring secure and efficient operations.

4.1.1 Technology Requirements

1. Technology Stack

- **Frontend:Frameworks:** Laravel Blade for templating, alongside Laravel Livewire for dynamic components.
- **InfyOm:** To streamline the development process with CRUD (Create, Read, Update, Delete) operations and scaffolding.
- **Backend:** PHP (Laravel framework)
- **Database:** MySQL

2. **Hosting:** The SMS will be hosted in the cloud, providing cost-effectiveness and scalability.
3. **Security:** Implementing security features such as encryption for sensitive data, multi-factor user authentication, and regular backups to ensure data integrity and security.
4. **Operating System:** The system will run on a server with a compatible OS, such as Ubuntu or CentOS, which are suitable for web applications and provide a stable environment for PHP and MySQL. During system use, the system will be multi-platform independent.
5. **Additional Requirements:** To optimize system performance and ensure seamless integration, WITI may need to consider:
 - **Upgraded Cloud Hosting Package:** Enhancing to a better cloud shared hosting plan to increase server capacity.
 - **Improved Internet Connectivity:** Upgrading internet services to support increased traffic and ensure reliable access.

Conclusion: The proposed Student Management System (SMS) is technically feasible, supported by WITI's existing infrastructure, including sufficient hardware, reliable Wi-Fi, and a capable cloud hosting server. Recommended upgrades to the cloud package and internet connectivity will enhance performance. Overall, the institution is prepared for a successful SMS implementation.

4.2 Operational Feasibility

Staff and Student Readiness: WITI's staff and students are prepared to adapt to the new Student Management System (SMS). Staff members are proficient in similar systems, while many have software development knowledge.

Integration into Current Workflows: The SMS will streamline tasks like registration, attendance tracking, and grading, reducing manual efforts and enhancing workflow efficiency.

Training Requirements

- **Staff:** No training needed as they are already skilled in terms of the use of such systems
- **Students:** Basic training sessions may be offered by staff to ensure familiarity with the SMS features.

Conclusion: Overall, the operational feasibility of implementing the SMS at WITI is strong, given the readiness of both staff and students, the positive impact on existing workflows, and the minimal training requirements. This readiness supports the successful adoption and long-term use of the proposed system

4.3 Economic Feasibility

4.3.1 Total Development Costs

1. Development Costs

- **Student Developers**

5 Best students working for 3 months.

Total = 5 students × UGX 300,000 × 3 months = **UGX 4,500,000.**

- **Professional Developer**

1 Proficient developer working for 3 months.

Total = UGX 2,000,000 × 3 months = **UGX 6,000,000.**

2. Infrastructure Costs

- **Cloud Hosting:** UGX 200,000 per year.

3. Testing Costs

- Since WITI staff can effectively test the functional features and provide feedback, this will reduce additional costs associated with third-party testing. Therefore, **UGX 0** for testing.

4. Maintenance Costs

- Estimated at **UGX 600,000** annually for the developer's maintenance support after deployment.

Total Estimated Project Costs = UGX 11,300,000

4.3.1 Cost-Benefit Analysis

Expected Benefits

- By leveraging the expertise of WITI's best students and a proficient developer, the institution minimizes development costs while providing valuable learning opportunities for students.
- The collaboration will enhance students' skills and practical experience while ensuring that the system aligns with WITI's needs.
- The anticipated benefits of increased operational efficiency, improved data management, and support for WITI's mission to bridge the technological gender gap will outweigh the development and maintenance costs.

Conclusion: The economic feasibility of the SMS at WITI indicates a viable project with costs totaling UGX 11,300,000. Utilizing existing resources and student developers minimizes expenses while enhancing efficiency and data accuracy. The anticipated benefits outweigh the costs, confirming the project's feasibility.

5.0 CONCLUSION

The feasibility study for the proposed Student Management System (SMS) at the Women's Institute of Technology and Innovation (WITI) supports its implementation across all dimensions.

Technical Feasibility: WITI has the necessary hardware, software, and network infrastructure, including cloud hosting, to support the SMS, which will utilize PHP (Laravel) and MySQL.

Operational Feasibility: Staff and students are ready to adapt to the new system, with proficient staff available to train students. Student developers will collaborate on the project, ensuring smooth integration into existing workflows.

Economic Feasibility: The estimated cost for development and maintenance is **UGX 11,300,000**. Utilizing existing resources and student involvement minimizes costs while enhancing efficiency and data accuracy.

In summary, the SMS project is feasible and aligns with WITI's mission to provide quality education and bridge the technological gender gap in Uganda.

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